Underspecification in RC attachment: A speed-accuracy tradeoff analysis

Pavel Logačev and Shravan Vasishth

University of Potsdam

pavel.logacev@gmail.com

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Overview

1. Two models of ambiguity resolution
   - Underspecification
   - Unrestricted Race Model (URM)
2. Experiment
   - Method
   - Predictions
   - Results
3. Asymptotic accuracy and attention

Ambiguity Advantage

Traxler et al. (1998): Eye-tracking experiment

(1) a. *high attachment*
The driver of the car [that had the *moustache*] was pretty cool.

b. *low attachment*
The car of the driver [that had the *moustache*] was pretty cool.

c. *globally ambiguous*
The son of the driver [that had the *moustache*] was pretty cool.

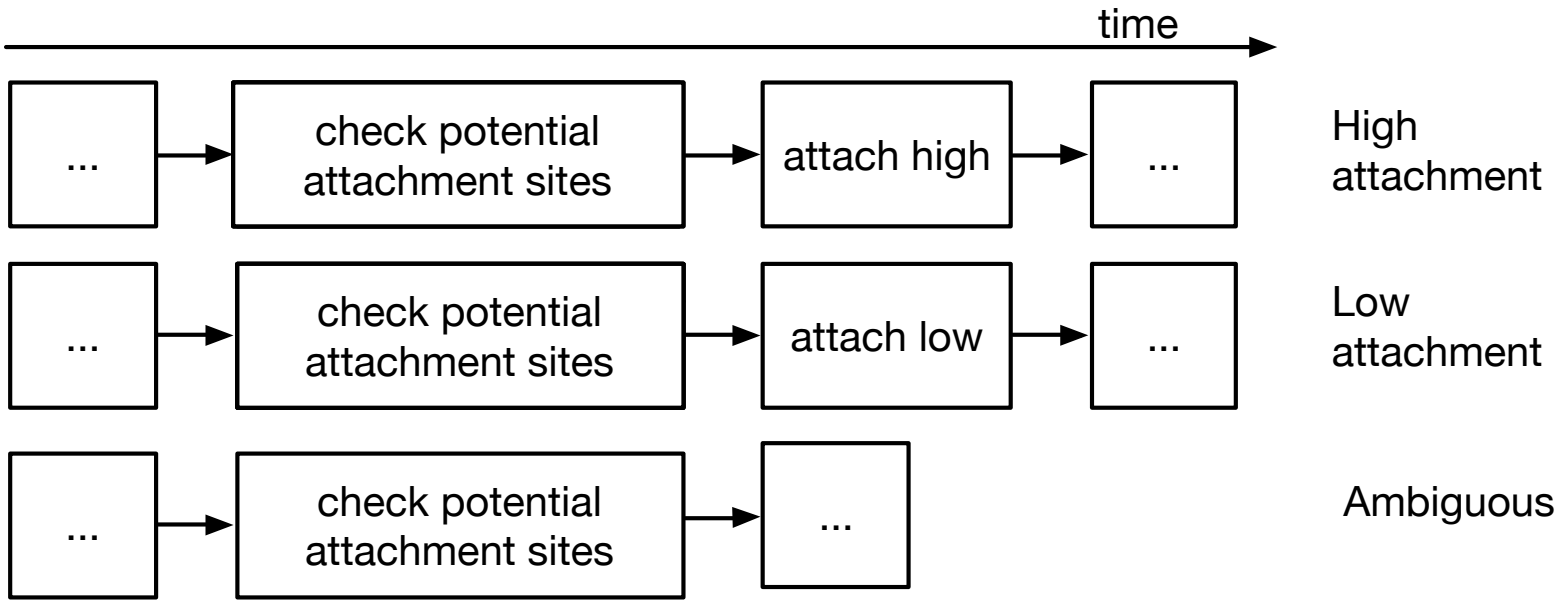
- *moustache* was read faster in condition (c) than in (a) or (b)
- several replications (e.g., van Gompel et al., 2001, 2005)

Underspecification

Swets et al. (2008): Underspecification

- When the parser detects an ambiguity, it disambiguates only if the task demands require it.
- One processing stage is omitted in ambiguous conditions.

→ Minimum processing time in ambiguous condition smaller than in unambiguous conditions.



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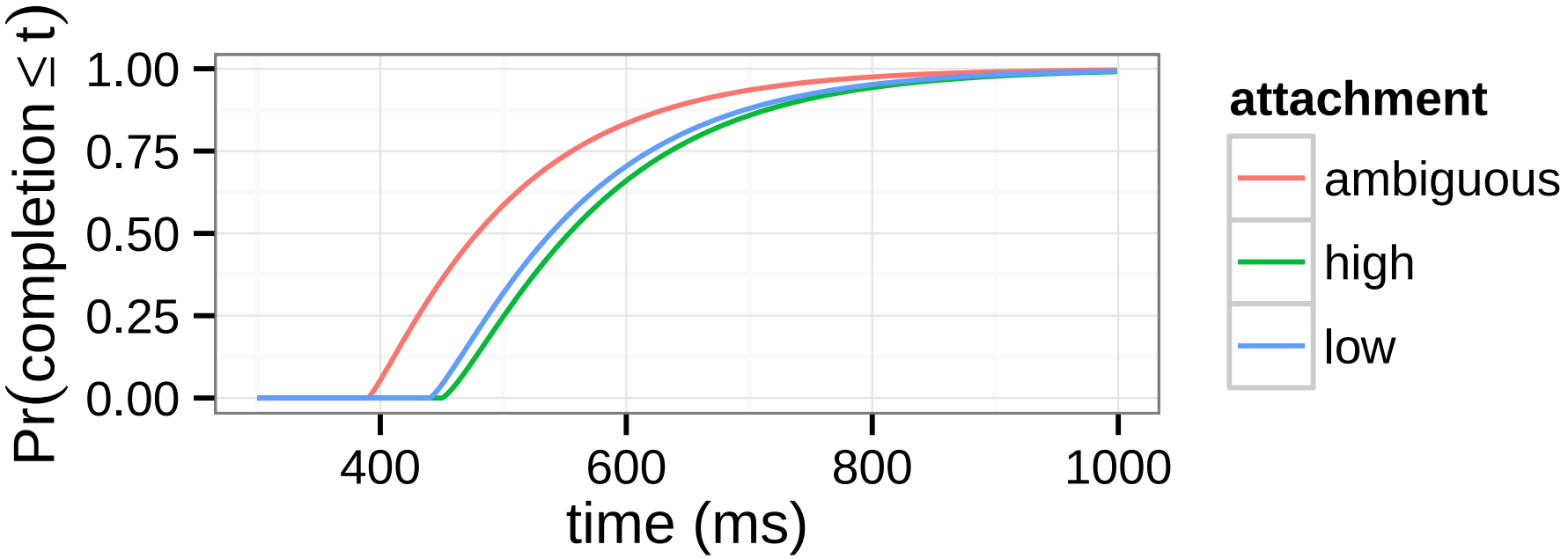

Unrestricted Race Model (URM)

Van Gompel et al. (2001): Unrestricted Race Model

- The parser tries to build two parses simultaneously.
- Whichever parse is constructed first, wins the race.
- Attachment completion times vary from trial to trial.
- Probability of finishing early is higher if two processes are involved, because their finishing times are independent.

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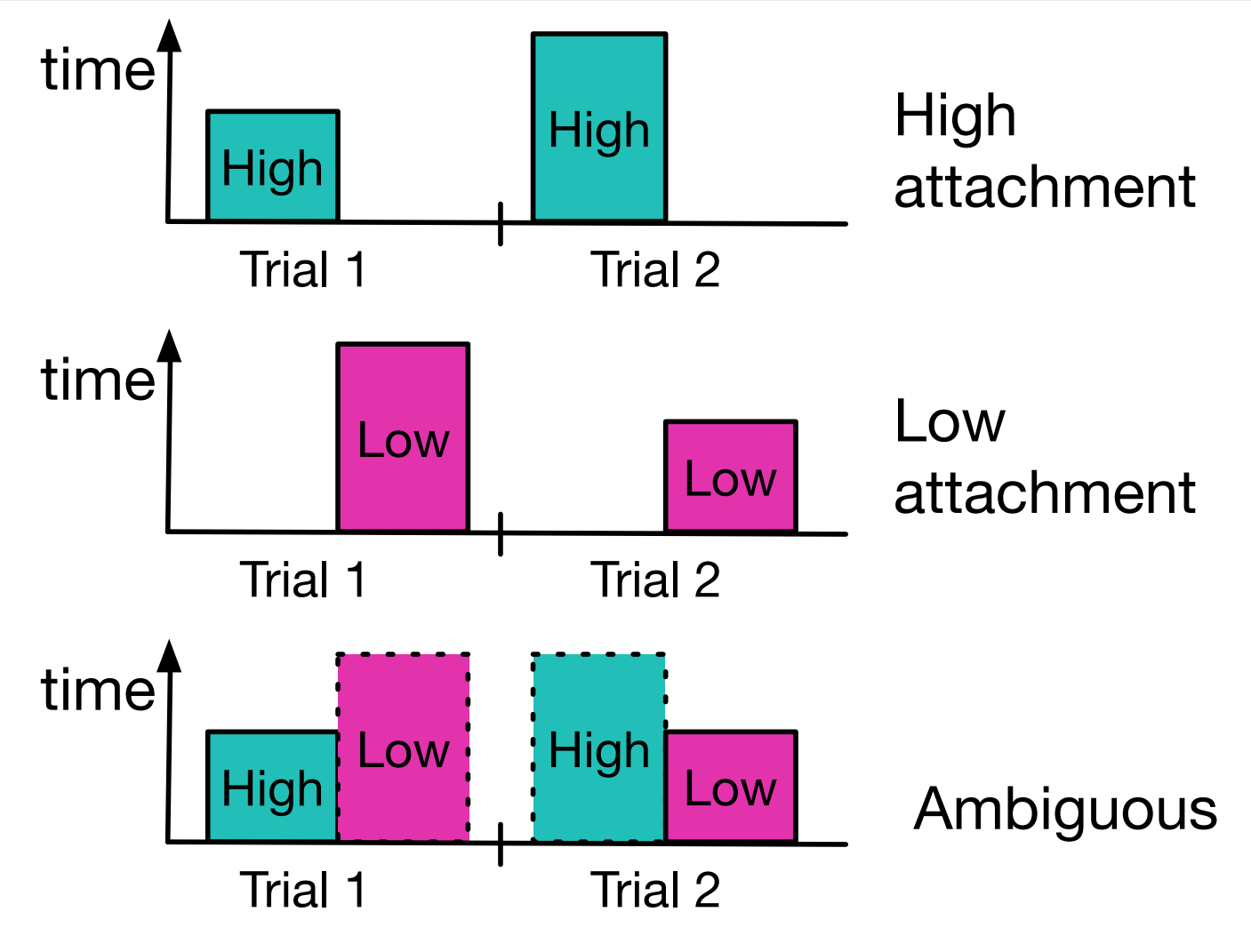

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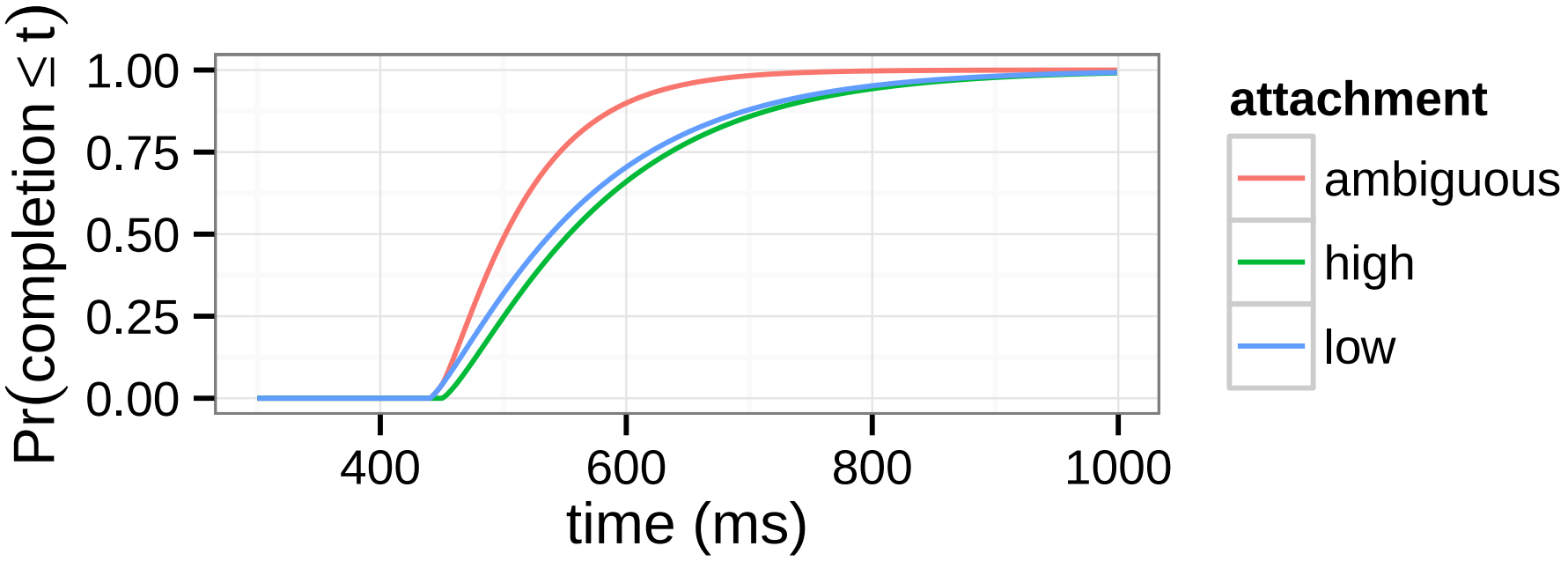


Experiment: Stimuli

a. [high]

Was	dachte	die	Managerin	des	Sängers,	die	schwieg?
What	thought	the	manager.FEM	of the	singer.MASC,	who.FEM	was silent

b. [low]

Was	dachte	der	Manager	der	Sängerin,	die	schwieg?
What	thought	the	manager.MASC	of the	singer.FEM,	who.FEM	was silent

c. [ambiguous]

Was	dachte	die	Managerin	der	Sängerin,	die	schwieg?
What	thought	the	manager.FEM	of the	singer.FEM,	who.FEM	was silent

d. *[ungrammatical]

Was	dachte	der	Manager	des	Sängers,	die	schwieg?
What	thought	the	manager.MASC	of the	singer.MASC,	who.FEM	was silent

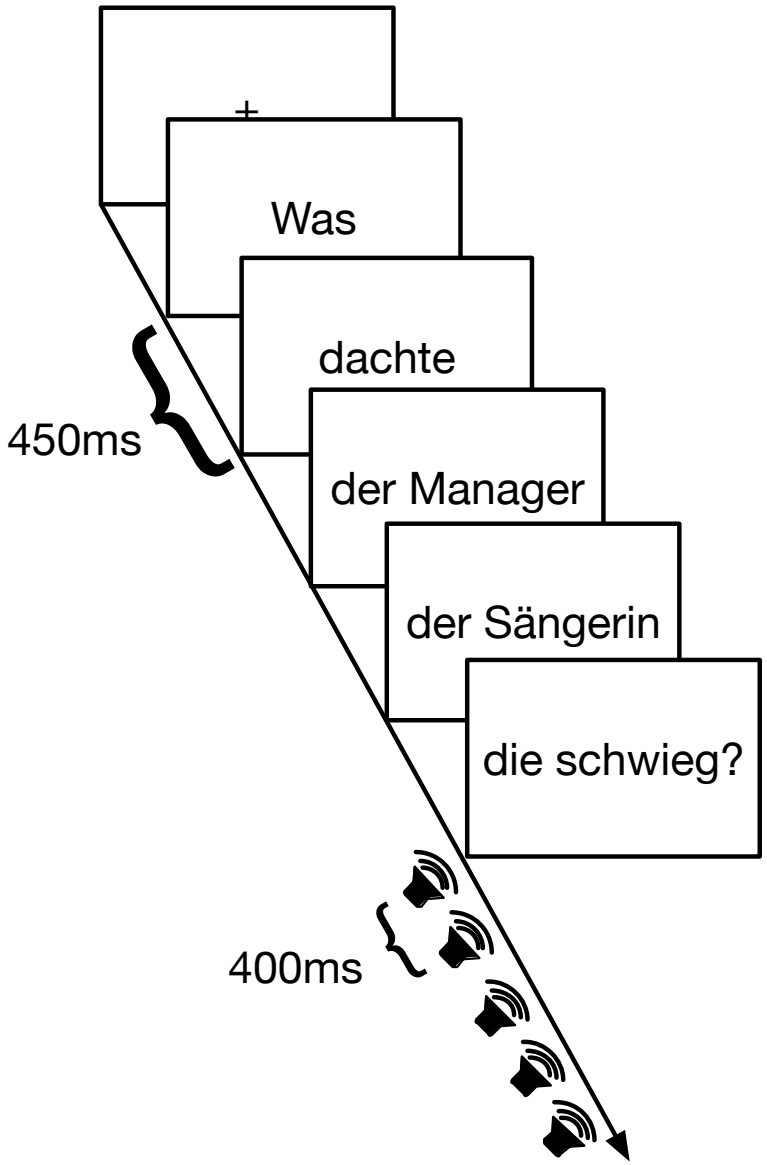
'What did the manager of the singer who was silent think?'

- Four further conditions, with masculine relative pronouns
- Thirty-two sentence sets

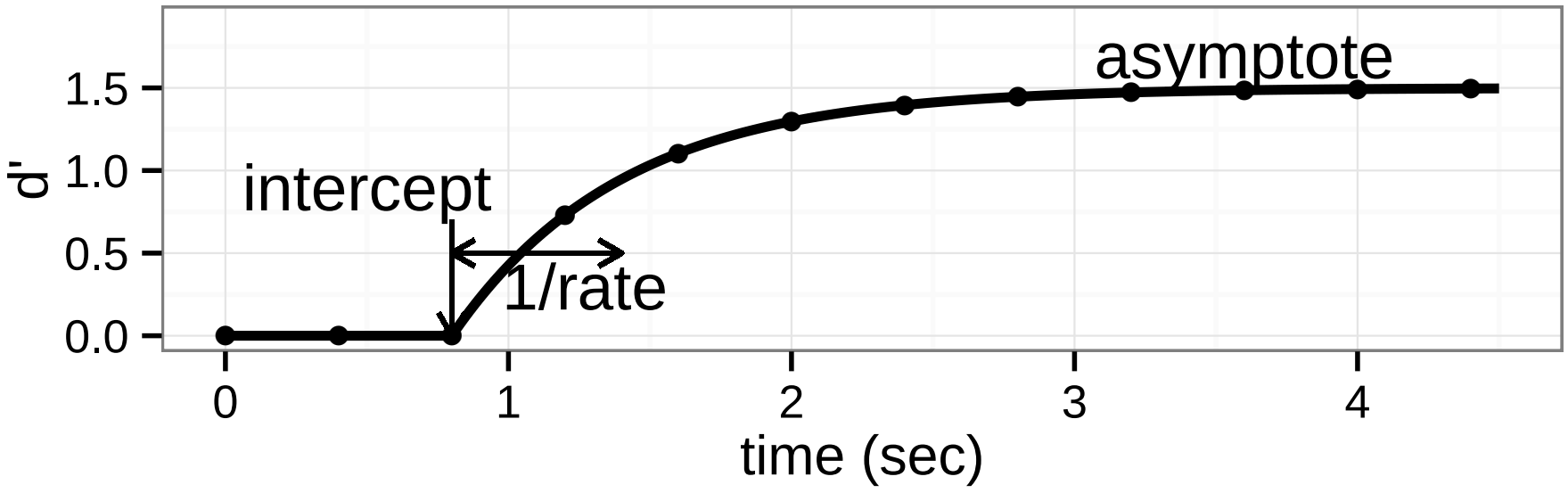
Experiment: Method

- Participants judged the acceptability of sentences at predetermined lags.
- 14 signal tones on every trial, with 400 ms SOA. (MR-SAT)
- We cannot measure the probability of successful completion, but we can measure accuracy

→ Bias-free measure of accuracy at different lags:
$d' = z(hits) - z(false\ alarms)$



Experiment: Method



- **Intercept** in d' reflects the earliest time at which processing *can* have finished
- **Asymptote** in d' reflects the level of accuracy achieved in a particular condition
- **Rate** in d' has no straightforward to the rate in %, because the relationship between % correct and d' is nonlinear

Experiment: Stimuli

- 20 participants (2 excluded).
- 192 acceptable experimental sentences (64 per attachment condition).
- 192 unacceptable sentences (unacceptable sentences repeated for counterbalancing).
- 128 filler sentences

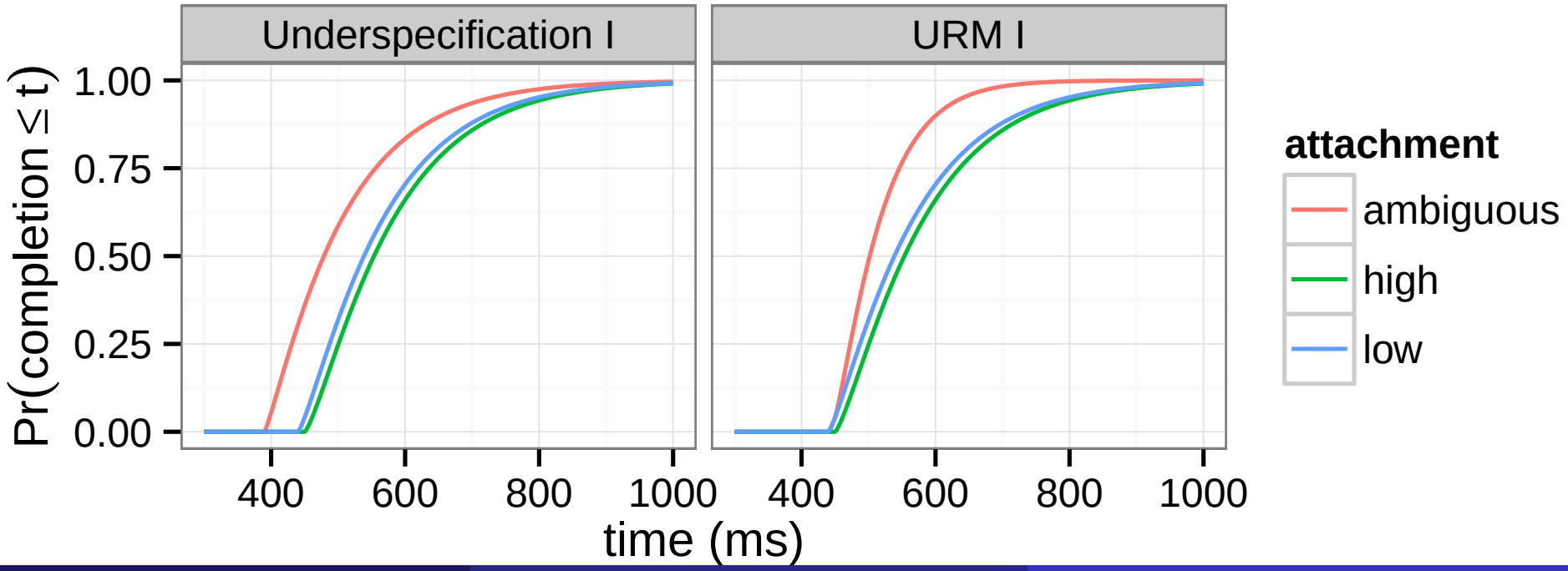
Experiment: Predictions

Underspecification

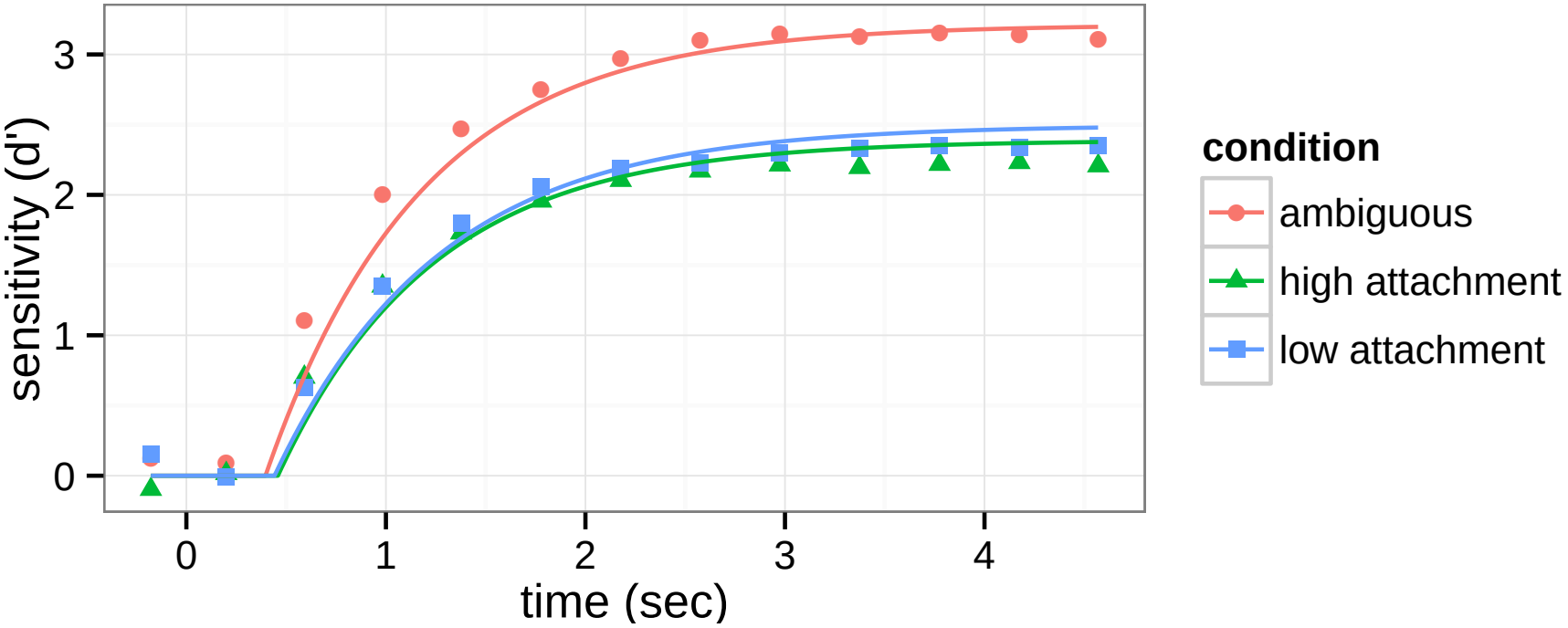
- Lower intercept in the ambiguous condition.

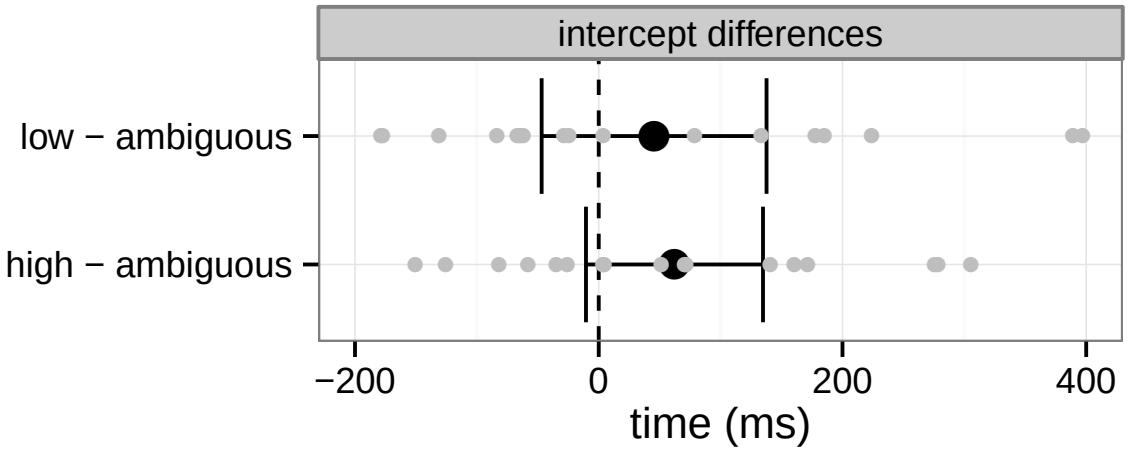
URM

- Intercept in the ambiguous condition equal to the unambiguous condition(s).



Experiment: Results

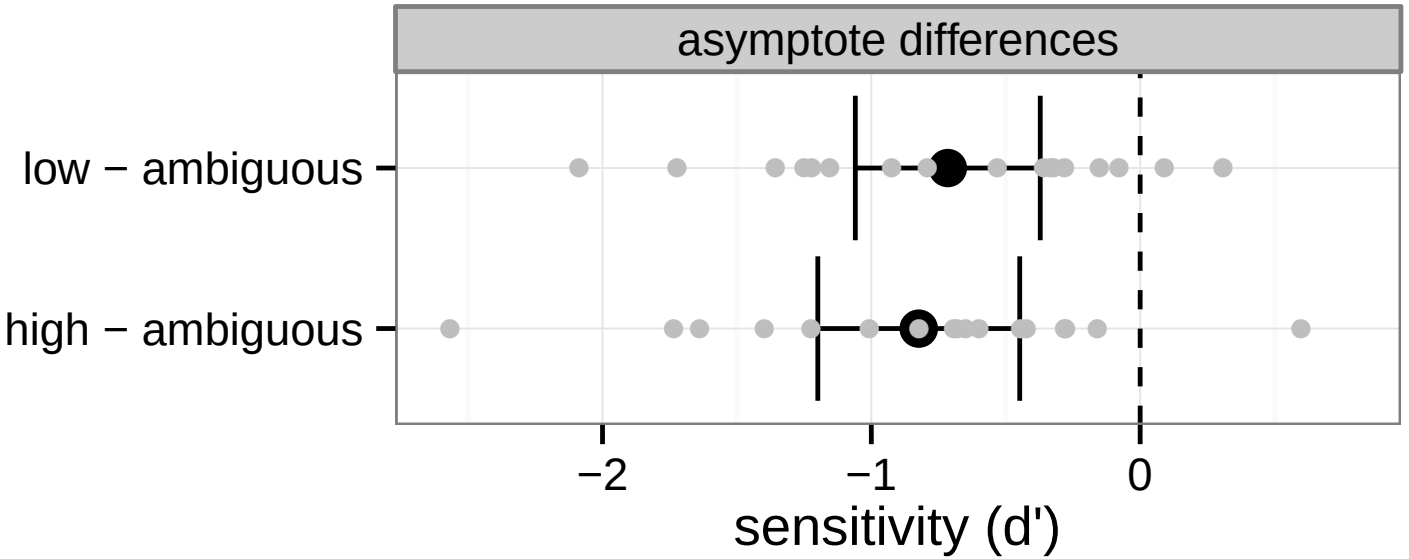




Experiment: Results

Accuracy at the latest lag

	ambiguous	high	low	none
% *acceptable*	0.91 (0.07)	0.75 (0.1)	0.8 (0.09)	0.11 (0.07)
d'	2.9 (0.1)	2.1 (0.1)	2.3 (0.1)	



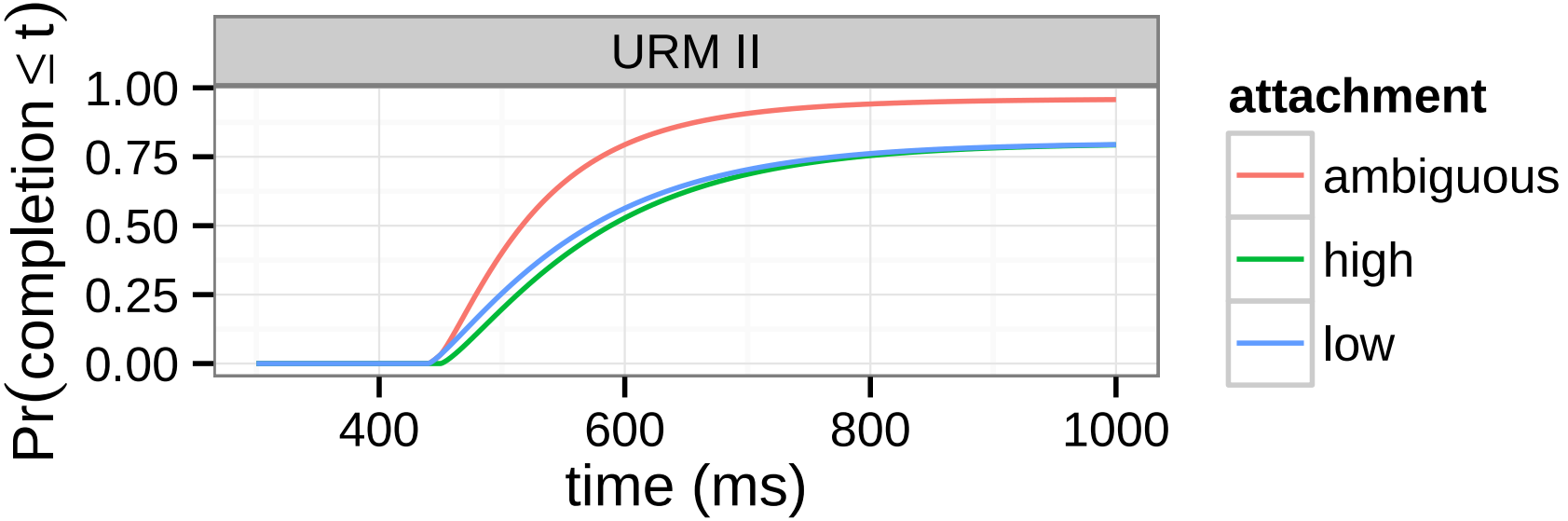
Discussion

- No significant differences in intercepts: No evidence for underspecification.
- Significantly higher asymptote in the ambiguous condition.
  - Higher asymptotes in SAT tend to correspond to faster reading in other paradigms.
  - → Higher asymptote seems to correspond to the ambiguity advantage.
- Model comparison on the basis of BIC: the same results

Asymptote differences in the URM

- RC attachment may be fallible (e.g., due to retrieval from memory)
- When retrieval of an NP fails, attachment fails and the respective attachment is considered impermissible.
- Assuming that each attachment process has a probability of failing:

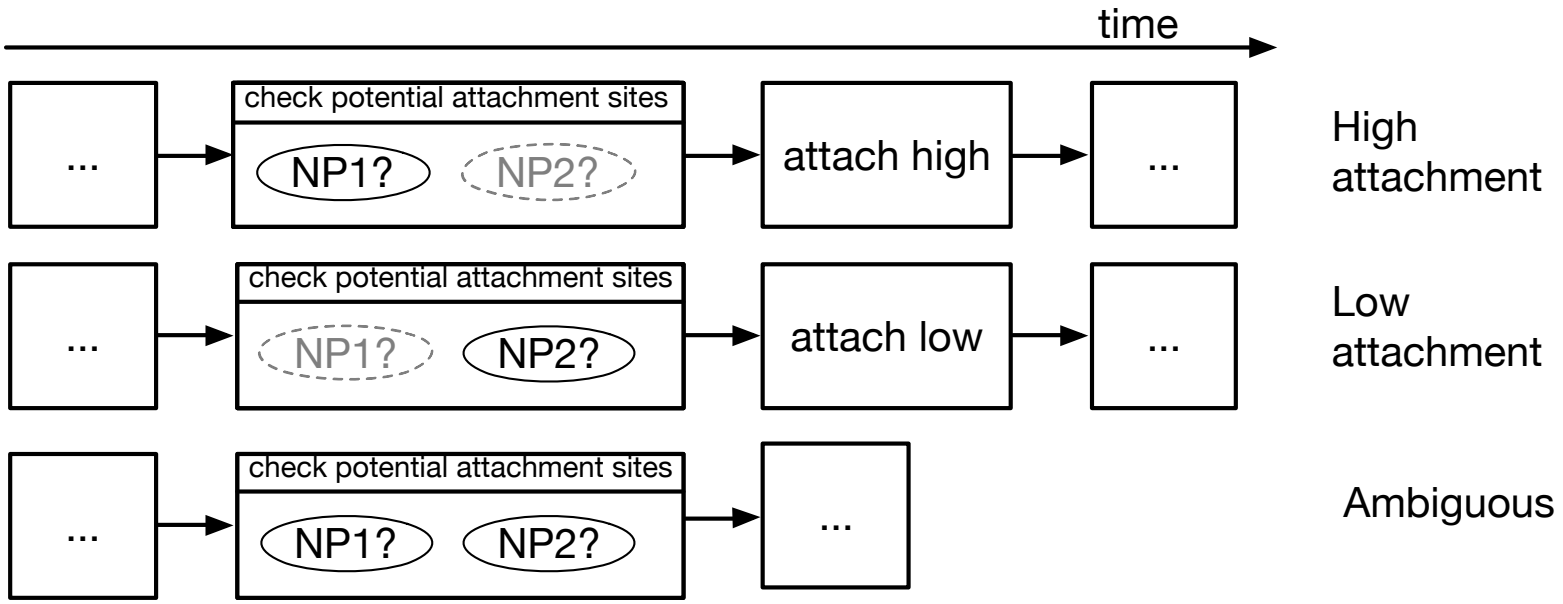
$$p_{fail_ambiguous} = p_{fail_high} \cdot p_{fail_low}$$



Asymptote differences in Underspecification

- Checking possibility of RC attachment may also be subject to failure.
- When retrieval of an NP fails, the parser could consider the corresponding attachment impermissible.

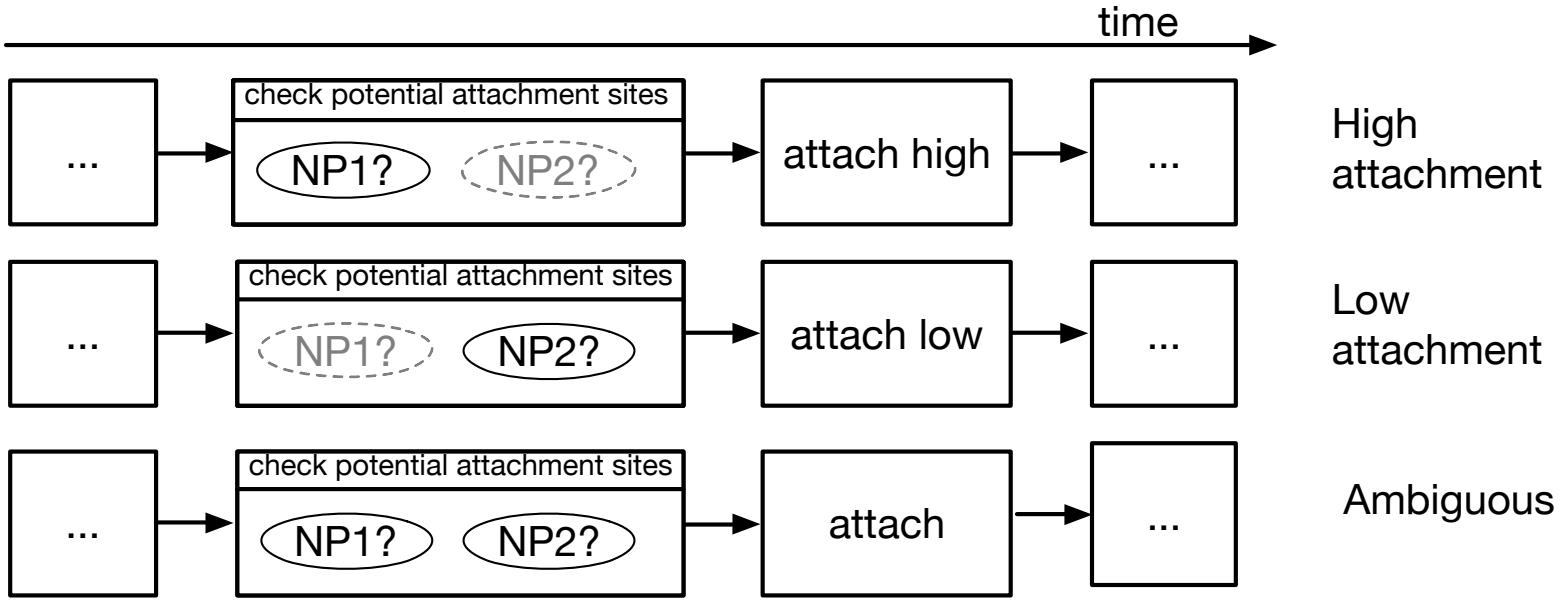
→ Predicts higher asymptotes in the ambiguous condition.



Underspecification does not need Underspecification

- Higher asymptotes tend to correspond to faster reading (e.g., Martin & McElree, 2008; Van Dyke & McElree, 2011).
- Predicted difference in asymptotes explains the ambiguity advantage.
- Underspecification requies a processing step (*checking attachment sites*), which can explain the ambiguity advantage.

→ Underspecification is not necessary to explain the speedup in reading.



Asymptotic accuracy and attention

	ambiguous	high	low	ungrammatical
% *acceptable*	0.91 (0.07)	0.75 (0.1)	0.8 (0.09)	**0.11 (0.07)**

Attention-loss I

- Participants sometimes 'zone out' for an entire trial, and try to guess the correct answer

→ Predicts equal numbers of attention-related errors in all conditions

Attention-loss II

- Because of our mode of presentation, participants may sometimes fail to process the gender of a noun-phrase (due to e.g., momentary lack of attention, or lack of time)

→ Predicts less errors in the ambiguous condition.

Asymptotic accuracy and attention

- Parameters for six models estimated by-participant.
- Models assuming that asymptote differences are attention-related have much worse fits.

asymptotes	attention	ΔBIC	logLik	n_{par}	$n_{selected}$
different	attention-loss I	0	-151	$3 \cdot 18$	8
different	attention-loss II	3	-152	$3 \cdot 18$	8
different	attention-loss I+II	72	-148	$4 \cdot 18$	-
same	attention-loss I	2823	-1639	$1 \cdot 18$	1
same	attention-loss II	901	-678	$1 \cdot 18$	-
same	attention-loss I+II	930	-654	$2 \cdot 18$	1

Conclusion

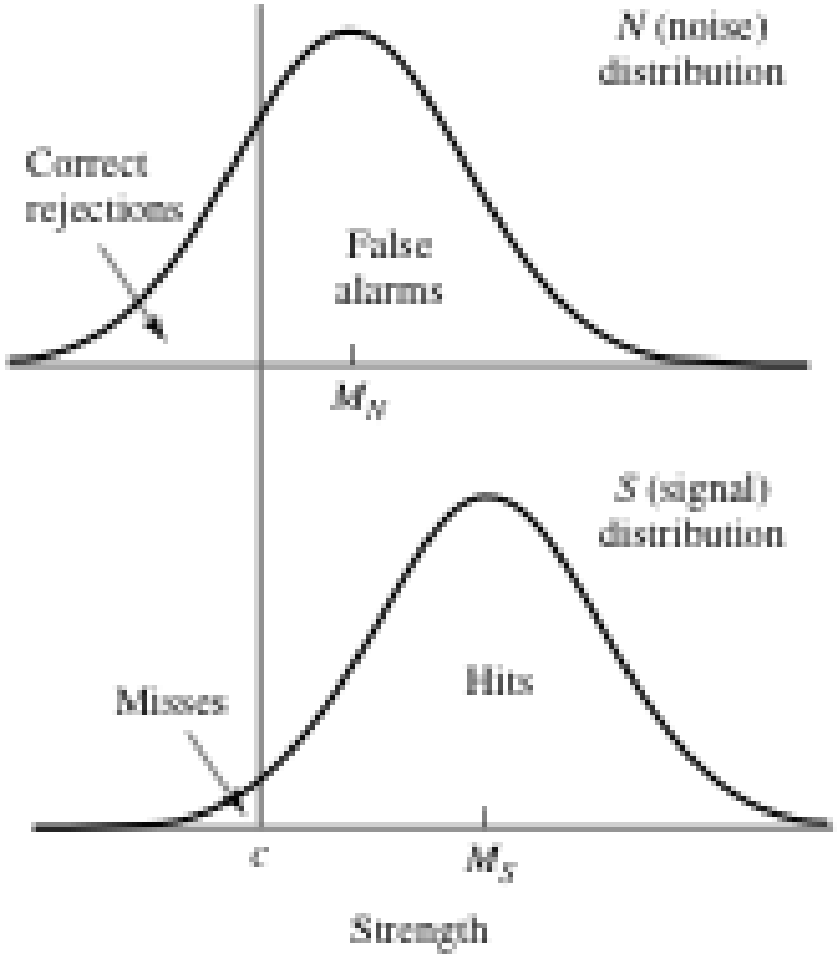
- The ambiguity advantage corresponds to asymptote differences.
  - Ambiguous RCs are processed more successfully than unambiguous RCs.
- No evidence for differences in intercepts predicted by underspecification.

Thank you.

d' as a measure of accuracy

Core assumptions of the signal-detection theory (SDT)

- The internal representation of a feature is affected by random (normally-distributed) noise.
- A 'yes'-response is made if the noisy representation is above a threshold (*criterion c*).
- *Sensitivity* d' is a bias-free measure of discrimination.



(Macmillan, 2000)

Parameter Estimation

- The change in sensitivity (d') and in decision criterion (c) was approximated by negatively accelerated exponential functions. Together, these two functions predict the proportions of hits and false alarms at any point in time.

$$d'(t) = asymptote_{d'}(1 - e^{-rate_{d'}(t - intercept_{d'})})$$
$$for\ t \geq intercept_{d'},\ otherwise\ d'(t) = 0$$

$$c(t) = asymptote_c(1 - e^{-rate_c(t - intercept_c)}) + initial_c$$
$$for\ t \geq intercept_c,\ otherwise\ c(t) = initial_c$$

- 64 models (with and without differences between conditions in intercept, rate, and asymptote) were fit to each parsticipant's data
- Maximum-likelihood estimation with a correlation parameter between successive data points datapoints within one trial

Model Comparison

$$BIC = -2 \cdot LL + log(n_{obs}) \cdot n_{par} \tag{1}$$

$$BF_{12} = \frac{P(D|H_1)}{P(D|H_2)} \tag{2}$$

$$BF_{12} \approx exp(\frac{1}{2}\Delta BIC_{21}) \tag{3}$$

$$GBF_{12} = \prod_k BF_{12(k)} \tag{4}$$

Model Comparison

- 64 models (with and without differences between conditions in intercept, rate, and asymptote) were fit to each parsticipant's data
- MLE with a correlation parameter between successive data points datapoints within one trial
- Bayesian information criterion (BIC) for model selection and the BIC approximation to the Bayes factor for inference (Wagenmakers, 2007)

$$BIC = -2 \cdot log(Lik) + n_{par} \cdot log(n_{obs}) \tag{5}$$

$$BF_{12} \approx exp(\frac{1}{2}\Delta BIC_{21}) \tag{6}$$

- Log-group Bayes Factor to aggregate evidence over participants (Stephan & Penny, 2006).

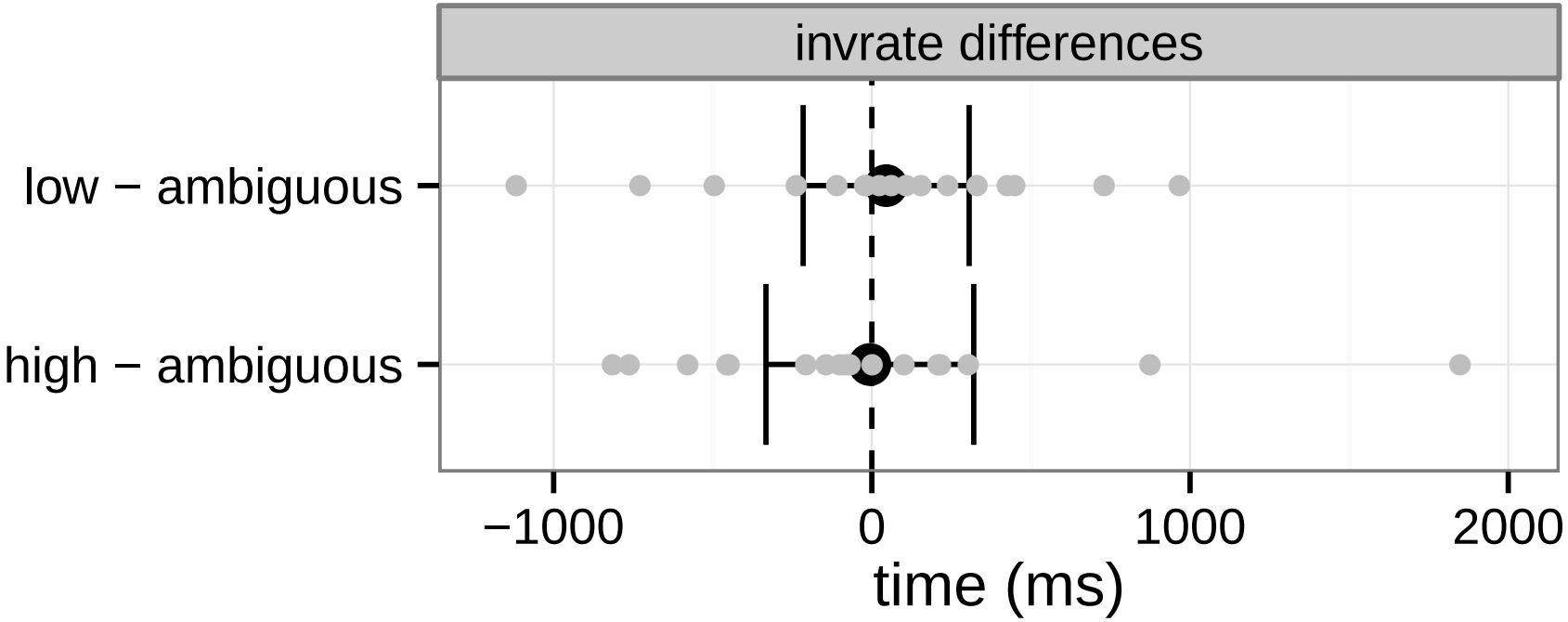
Model Comparison: Best-ranked models according BIC

	asymptote		1/rate		intercept					
rank	high	low	high	low	high	low	ΔLL	ΔBIC	$LGBF_1$	n_{par}
1.	-0.8	-0.7					0	0	0	5
2.	-0.5						-104	48	50	4
3.			1307				-133	107	79	4
4.							-218	119	111	3
5.	-0.5		-112				-61	121	61	5
6.	-0.8	-0.7	-64				10	138	43	6
7.	-0.6			613			-80	160	80	5
8.	-0.4				47		-83	166	83	5
9.				336			-166	173	112	4
10.	-0.6			682	80		-9	177	63	6
42.			1337	576	64	99	-67	453	175	7

- ΔBIC: the model's BIC minus the smallest BIC.
- ΔLL: the model's log-likelihood minus the log-likelihood of the model selected according to BIC.
- LGBF values above 3 and 5 can be considered strong and very strong evidence, respectively.

Model Comparison: By-participant

- Numbers of participants for which particular parameterizations were selected on the basis of BIC.
- Brackets: Numbers of participants with a preference supported by a Bayes factor of 3 or more.

contrast	asymptote	rate	intercept
1	7 (6)	15 (11)	16 (12)
2-high	4 (3)	1 (1)	0
2-low	1 (1)	2 (2)	2 (2)
3	6 (5)	0	0

Results: 1/rate Differences